nature communications

Article

https://doi.org/10.1038/s41467-026-71206-4

Deep homology and design of proteasome chaperone proteins in *Candidozyma auris*

Received: 20 August 2025

Accepted: 17 March 2026

Published online: 30 March 2026

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A central tenet of biology is that protein structure mediates the sequence-function relationship. Recently, there has been excitement about the promise of advances in protein structure modeling to generate hypotheses about sequence-structure-function relationships. Here, we leverage structural similarity to identify rapidly evolving proteasome assembly chaperones and characterize their function in *Candidozyma (Candida) auris*. Despite extensive sequence divergence, we demonstrate conservation of function, corroborating that specific folds, and not sequences, are required for function. This theoretical premise suggests that protein structures with certain properties should be functionally interchangeable, even if they were not products of a common evolutionary history. To reduce this theory to practice, we performed structure-informed protein design, exploring sequence space that is not accessible via stepwise evolution, and mutated more than 40 residues in the Poc4 proteasome assembly chaperone to demonstrate that artificial proteins can rescue complex biological processes in the context of the whole cell. This sequence-structure-function relationship expands our ability to use structure to identify deep evolutionary relationships between proteins and generate hypotheses about gene function in non-model organisms. Overall, this helps to define and understand functional constraints on protein evolution, with important implications for both future protein design and retrospective function prediction.

The causal relationship between sequence, structure, and function is central to our ability to assign function for newly discovered proteins and explore evolutionary variation based on inferred homology. While the importance of BLAST and other sequence similarity searches cannot be overstated, these approaches can fail to detect evolutionarily related proteins with high levels of sequence divergence[1]. However, since structure mediates the sequence-function relationship, signatures of shared evolutionary history are often better preserved at the level of structure than sequence[2–4]. These structurally-related but sequence-distant proteins are often considered to be in the "twilight zone"[1,4,5]. Historically, homologous relationships of such proteins were difficult to identify at scale due to the challenges of protein structure determination, leading to their exclusion from functional genomic analyses. However, excluding proteins with low sequence identity can result in not just incomplete but systematically biased inferences about the composition and evolution of homologous systems. As an example, co-evolution analyses require an estimation of members of an orthogroup before calculating the evolutionary rate[6]. If rapidly

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https://doi.org/10.1038/s41467-026-71206-4

evolving proteins are not included, the inferred identity and number of the coevolving units, and their respective rates of evolution, would be artefactually depressed.

Recent advances in computational protein structure prediction have enabled large-scale comparative structural analyses of proteins across broad swathes of evolution[3,7–10]. In turn, it has become possible to experimentally probe the constraints and flexibilities within which sequence-structure relationships evolve[11–13]. Studies of natural protein sequence variation show that multiple protein sequences—homologous or convergently evolved—often result in equivalent structures and thus similar function (Fig. 1A, points A to B). We can also examine functionally equivalent proteins to develop hypotheses about the structural requirements for function. In principle, such discoveries from testing such hypotheses should of de novo designed proteins that are imbued with the structural features needed to fulfill the biological requirements of the cell, even though they are disconnected from evolutionary history (Fig. 1A, points B to C)[14].

Here, we explicitly test whether rapidly evolving protein-coding sequences predicted to encode similar structures result in similar functional phenotypes across evolutionary time. Such cases of structural conservation amidst sequence divergence are often viewed as cases of neutral evolution, where random sequence changes can accumulate between species because they do not result in the structural/functional changes that can then be selected for or against. As a representative case study, we examine components of proteasome assembly. Despite being involved in an essential, ubiquitously important process, we find that these assembly chaperone proteins vary considerably in their rate of evolution. In the non-model human fungal

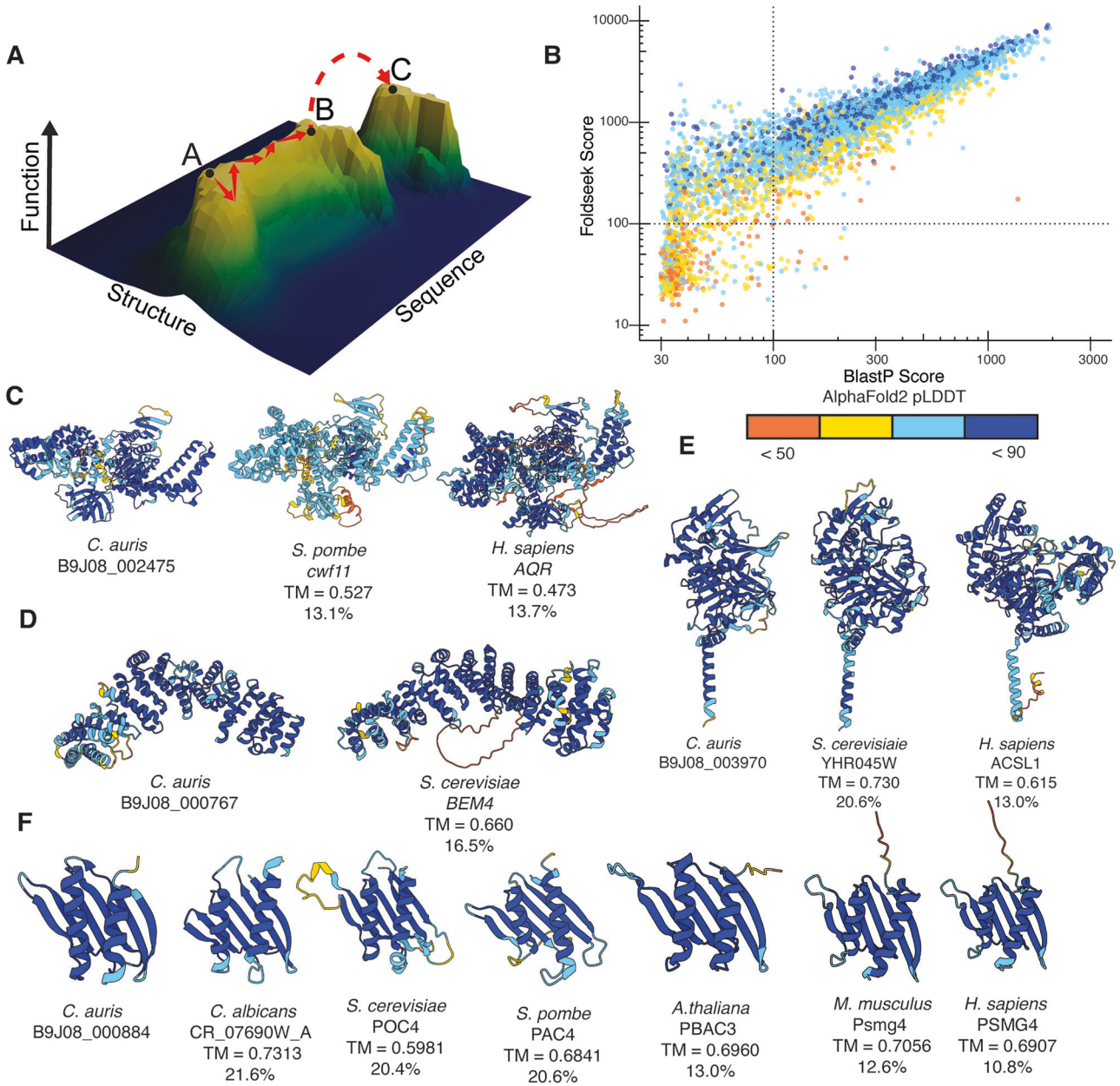


Fig. 1 | Structural homology can be used to predict protein function in the absence of sequence homology. A Diagram of a hypothetical fitness landscape, with the X axis as sequence, Z axis as structure, and Y axis as fitness. **B** Detection of remote homology of *C. auris* proteins by structure (Foldseek with AlphaFold Database predicted structures) vs. sequence (BlastP). For each of the 5008 proteins for which there is an AFDB structure, a plot of the best bit-scores to a protein in a Swiss-Prot proteome colored by the pLDDT, where the 100-bit threshold is indicated. **C**–**F** AlphaFold 2 models of *C. auris* proteins and structure matches identified by Foldseek for B9J08_002475 (**C**), B9J08_000767 (**D**), B9J08_0003970 (**E**), and B9J08_000884 (Poc4) (**F**). Models are colored by pLDDT by residue. Percentage indicates amino acid identity to the *C. auris* protein.

pathogen *Candidozyma (Candida) auris*, we identify and empirically validate a proteasome assembly chaperone that was previously undetectable by sequence because of its rapid evolution, but which retains structural signatures of homology. Disruptions of this protein in *C. auris* yield the same phenotype as disruption of its homologous counterparts in *Saccharomyces cerevisiae*, suggesting that the conservation in structure is sufficient to preserve function in proteasome assembly. Complementation of the *C. auris* mutant with the *S. cerevisiae* protein is sufficient to rescue growth, despite extensive sequence divergence. As this suggests that the folded structure of this protein may be sufficient to confer function, we use deep learning to design artificial proteasome chaperones to maintain the conserved structure and use these to rescue the *C. auris* mutant phenotype. This allows us to explore protein sequence space that was otherwise inaccessible due to epistatic constraints that would occur during stepwise evolution[15,16]. We then analyze the designed proteins to better characterize the structural determinants of function and the sequence determinants of structure. Our work demonstrates how modern computational methods, in combination with experimental characterization, can be used to explore the evolution of sequence-structure-function relationships and, from the insights gained, enable us to create functional molecules released from the constraints of evolutionary history.

Results

Structural homology in the absence of sequence homology predicts protein function

Structure-based similarity search is recognized to outperform sequence-based homology in retrospective benchmarks, and therefore structure-based homology search should enable better inferences of gene functions across large evolutionary distances where sequence-based methods fail[17–21]. These structure-based searches have been applied to specific classes of proteins but have not been fully explored at the genome scale, especially in *C. auris*. To examine this relationship, we compared the predictive power of Foldseek, a structure-based homology search algorithm, to BlastP, a sequence-based homology search algorithm, to detect orthologs for *C. auris* proteins amongst the proteins in the Swiss-Prot database (Fig. 1B, Supplementary Fig. 1A). As expected, we observed a high correlation between their respective bit-scores over the 5008 genes that had predicted AlphaFold2 structures (Spearman Rank Correlation: 0.932, p-value < 0.0001). However, we identified 204 genes where structural homology may be able to predict function in the absence of sequence homology, as determined by significant Foldseek E-values and insignificant BlastP E-values. For example, the protein B9J08_002475 and its sequence orthologs are *Metschnikowiaceae*-lineage restricted (Fig. 1C) but have significant structural homology with the Aquarius family of RNA helicases that function in the U2 snRNP spliceosome (Supplementary Table 1) despite sharing less than 15% sequence identity. Similarly, B9J08_000767, which is a *Candida*-lineage restricted protein, has significant structural similarity to the Bem4 guanidine exchange factor in *S. cerevisiae* but only 16.5% sequence identity (Fig. 1D, Supplementary Table 2). We are also able to leverage Foldseek against uncharacterized proteins present broadly throughout fungi, including B9J08_003970, which has structural similarity with mammalian long-chain fatty acid acyl-CoA synthetases 1, 5, and 6 (Fig. 1E, Supplementary Table 3).

During evolution, selection acts not just on single proteins but the entire functional unit of a protein complex[6,22]; therefore, we were particularly interested in the case of two proteins putatively involved in the highly coordinated and conserved process of proteasome assembly. We identified the uncharacterized protein B9J08_000884, with a similar structure to the Poc4 assembly chaperone protein, and B9J08_003917, which is annotated as a homolog of the Irc25 proteasome assembly chaperone (Supplementary Table 4)[23–25]. We will refer to these two proteins as Poc4 and Irc25, following nomenclature from *S. cerevisiae*. Proteins with structural similarity to Poc4 were widely present throughout Eukarya, and despite sequence identities below 20%, the overall topology of the protein domains is predicted to be conserved (Fig. 1F, Supplementary Fig. 1B). As the proteasome itself is highly conserved throughout Eukaryotes and has been implicated in human diseases such as neurodegeneration and cancer[26], we decided to investigate the evolution and conservation of this assembly machinery further.

Sequence divergence in proteasome assembly chaperones

To better understand how distinct components of the proteosome might have evolved, we used amino acid and syntenic analysis to search for homologs of proteosome assembly components across fungi. The putative orthologs of Poc4 and Irc25 function as a heterodimer and mediate assembly of α-subunits into the heptameric ring structure of the proteosome, interacting especially with the α5-subunit (Pup2), before dissociating during β-subunit assembly (Fig. 2A)[26]. The Pba1-Add66 complex also contributes to α-ring assembly around the Ump1 chaperone. We found that while the Pup2 proteasome subunit and the Ump1 assembly chaperone had detectable homologs throughout *Saccharomycotina*, the CTG clade lacked a clearly identifiable sequence that resembled the Poc4 proteins found in other yeast, and putative homologs of the other proteasome chaperones were at times only identifiable because of synteny (Fig. 2B). The similarity in structure between Irc25 and Poc4 and the human proteasome assembly chaperone protein PSMG3 also brings up the possibility that Poc4 and Irc25 are highly diverged paralogs.

The variable presence of different proteosome components suggested that the folding chaperones and core components may have different evolutionary rates from each other. To compare the extent of divergence among core proteins and folding chaperones, we used the putatively orthologous sequences from BlastP and inferred the maximum likelihood phylogeny describing the evolutionary history of each gene using the best-fit model of sequence evolution, as determined by Akaike information criterion (Supplementary Table 5, Supplementary Fig. 2A–C). The proteins investigated displayed vastly different rates of divergence, with the assembly chaperones showing rapid evolution. As expected, the evolutionary distance between Pup2 orthologs is much smaller than the difference between chaperone orthologs (Fig. 2C, E). The lack of mating and gene flow in the CTG clade can allow for rapid drift. However, this is not the only cause of rapid evolution of the chaperones as the higher rate of evolution among the chaperone proteins is not unique to any species lineage but is true across all taxa (Fig. 2D, E, Supplementary Fig. 3). Importantly, we also observe that Poc4 appears to be evolving faster than its co-chaperone Irc25 within the CTG clade (Fig. 2D, E, Supplementary Fig. 3D). One possible reason for this discrepancy in chaperone evolution rate is expression levels, as higher expression is generally associated with lower evolutionary rate[27]. Previously published RNA-seq data show that Poc4 expression is much lower than other core particle assembly chaperones (Supplementary Fig. 4)[28], but the translation rate is not known. To examine the possibility that *POC4* is in a specifically mutagenic locus, we examined the sequence divergence of genes near *POC4*. We failed to observe high rates of sequence divergence among genes near the *POC4* locus, suggesting that this divergence is not due to its presence in a mutational hotspot (Supplementary Fig. 5).

C. auris POC4 encodes a bona fide proteasome assembly chaperone

While we observed structural similarity to known Poc4 proteasome assembly chaperones, the extreme sequence divergence suggested that its role in proteasome assembly might not be preserved. Furthermore, while previous studies have identified structural conservation in the absence of sequence identity across large evolutionary distances, many of these predictions lack direct experimental evidence

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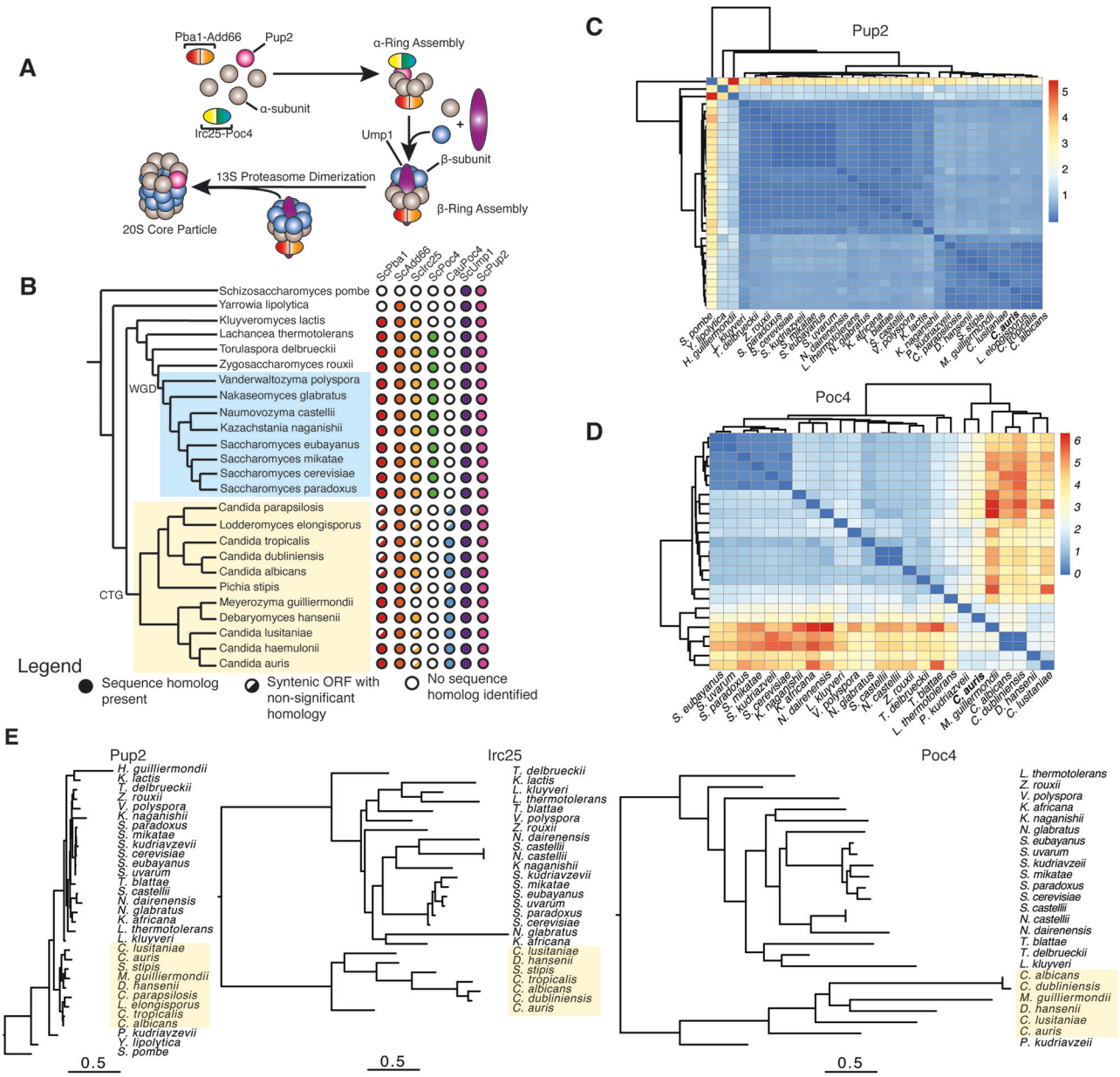


Fig. 2 | *C. auris* Poc4 is sequence divergent, but not structurally divergent, from known Poc4 assembly chaperones. A Schematic of 20S core particle assembly. The Pba1-Add66 and Irc25-Poc4 complexes initiate assembly of α-subunits into a heptameric ring structure, where Irc25-Poc4 directly interacts with the Pup2 subunit. Before β-subunit incorporation, Irc25-Poc4 dissociates, and Ump1 enters the central cavity to create a half proteasome. Two half proteasomes dimerize to create a 20S core particle. **B** Evolutionary relationship of analyzed yeast species shows a rapid loss of sequence identity among proteasome assembly chaperones. Sequences with similarity to the indicated *S. cerevisiae* and *C. auris* proteins were identified using BlastP from NCBI using an E-value cutoff of 10^{-5} and requiring hits to reciprocal best matches. CTG indicates the clade where the CUG codon is translated as serine instead of leucine. WGD denotes the whole genome duplication clade. **C**–**D** Pairwise comparisons of maximum likelihood distances between Pup2 (**C**) and Poc4 (**D**) sequences. **E** Evolutionary rate of Pup2, Irc25, and Poc4 sequences among the indicated yeast species identifies a rapid change in chaperone proteins. Scale = 0.5 substitutions per site.

regarding the conservation of protein function. Proteins with highly conserved folds often display significant differences in catalytic activity and substrate specificity[29,30]. Therefore, to test whether *C. auris* Poc4 retained canonical interactions with the proteasome, we performed co-immunoprecipitation assays with an ALFA-tagged variant of Poc4 and the α-subunits of the 20S core particle. We found this physical interaction was preserved (Fig. 3A and Supplementary Fig. 6A, B), suggesting that the capacity to act as an assembly chaperone had persisted in Poc4 even as its amino acid sequence changed beyond recognition.

Next, we directly characterized whether this physical interaction was involved in the same cellular functions. Disruption of proteasome function through genetic perturbations (e.g., deletion of Pre9 subunit) or pharmacological inhibition (e.g., MG-132 treatment) increased the level of ubiquitinated proteins within a cell (Fig. 3B). We reasoned that disruption of Poc4 should also produce these phenotypes. Consistent with our model, we observed increased total ubiquitinated proteins in Poc4 and Irc25 deletion strains relative to the wild-type AR0382 strain (Fig. 3B).

MG-132 treatment had limited effects on fungal growth at permissive temperatures but significantly reduced growth at high temperatures across three fungal species (Fig. 3C), demonstrating a conserved connection between proteasome inhibition and thermotolerance across organisms. Similar results were observed with a 2nd

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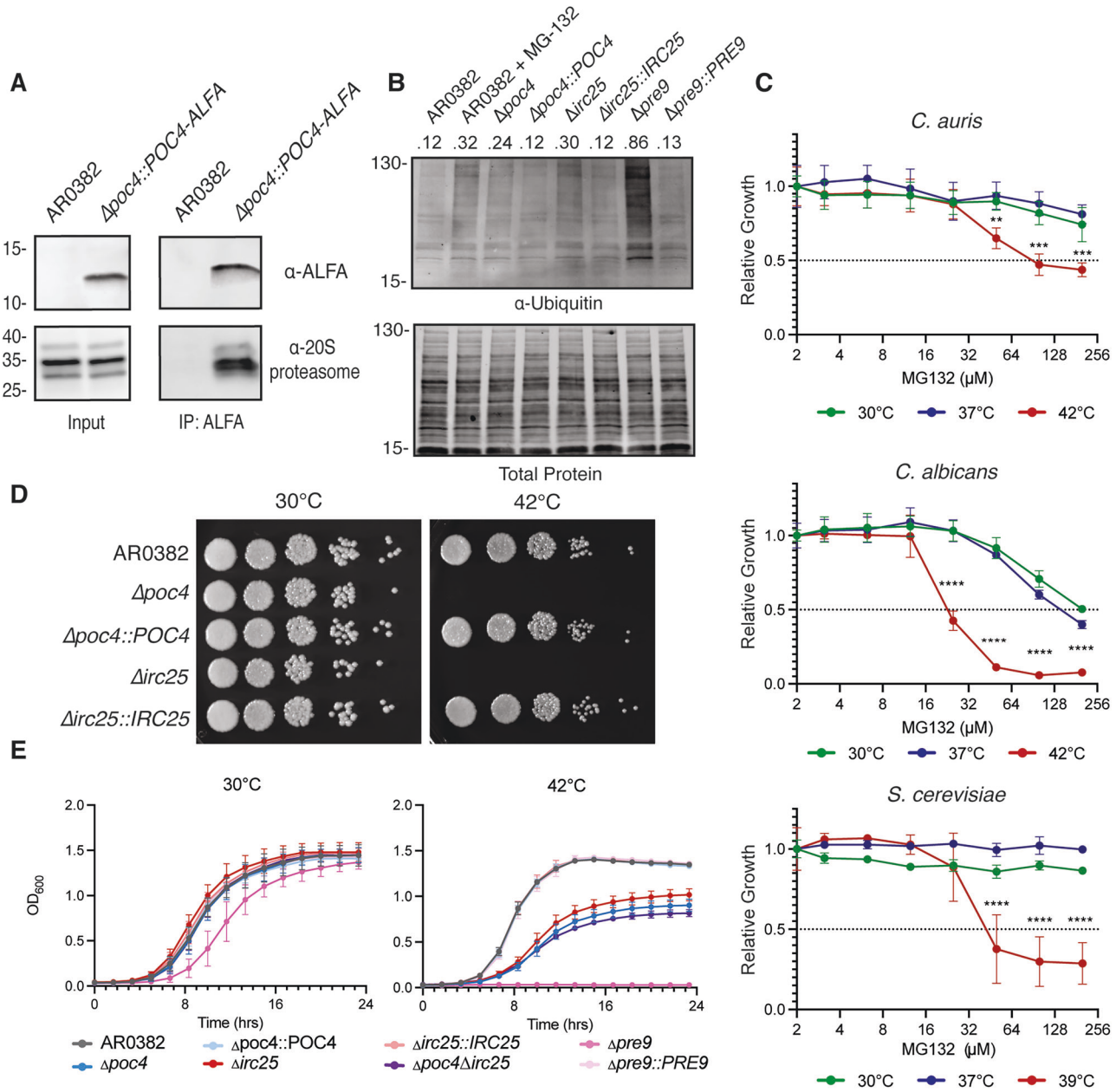


Fig. 3 | *C. auris POC4* encodes a bona fide proteasome assembly chaperone. A *C. auris* Poc4 physically interacts with the 20S proteasome. Immunoprecipitation of Poc4-ALFA with anti-ALFA co-purifies proteasome α-subunits, which do not co-purify with untagged controls. Numbers on left of blots indicate approximate size in kDa. **B** Total protein ubiquitination increases in *C. auris* upon loss of *POC4* or *IRC25*. Numbers above lanes indicate relative Ubiquitin intensity normalized to total protein intensity. **C** Proteasome inhibition reduces fungal growth at high temperatures. Dose-response curves of *C. auris*, *C. albicans*, and *S. cerevisiae* to MG-132 at the indicated temperatures. OD_{600} at each drug concentration is normalized to the growth of each strain in the absence of drug to determine relative growth. Data show mean ± standard deviation across 3 biological replicates. **D** Poc4 and Irc25 are both required for growth at 42 °C. Representative spot plates of *C. auris* grown at 30 and 42 °C. Indicated strains were diluted 10-fold and plated onto YPD agar and incubated for 48 h before imaging. **E** Growth curves of indicated *C. auris* strains at 30 and 42 °C. Growth was measured by OD_{600} every 10 min for 24 h. Data shown are the mean ± standard deviation across 4 biological replicates every 80 min for clarity. Statistical significance among different groups was calculated using two-way ANOVA, with Dunnett's post hoc tests for multiple comparisons and specific *p*-values for all comparisons are listed in Source Data. $^{*}p \le 0.05$; $^{**}p \le 0.01$; $^{***}p \le 0.001$; n.s., $p > 0.05$.

proteasome inhibitor, Bortezomib, but not general antifungal drugs (Supplementary Fig. 7A, B). Furthermore, *C. auris* cells lacking Poc4 or Irc25 are hypersensitive to MG-132 even at permissive temperatures (Supplementary Fig. 7C). Consistent with the pharmacological data, loss of either *IRC25* or *POC4* resulted in a temperature sensitive phenotype, where both mutants are unable to grow at 42 °C, but growth is restored upon complementation (Fig. 3D). Additionally, the *Δirc25* and *Δpoc4* individual deletion strains phenocopy each other, but deletion of both genes does not alter the phenotype beyond a single deletion, suggesting non-redundant functions for these proteins (Fig. 3E). We also do not see differences in protein expression of either Poc4 or Irc25 upon deletion of the reciprocal co-chaperone, unlike murine *PAC3* and *PAC4* expressed in HEK293T cells[23] (Supplementary Fig. 8). By contrast, deletion of the gene encoding Pre9, a proteasome subunit, results in temperature sensitive phenotype even at 37 °C (Supplementary Fig. 9A, B). Taken together, these experimental data demonstrate that, despite losing detectability as a sequence homolog, *C. auris* Poc4 has retained a conserved evolutionary role in proteasome function.

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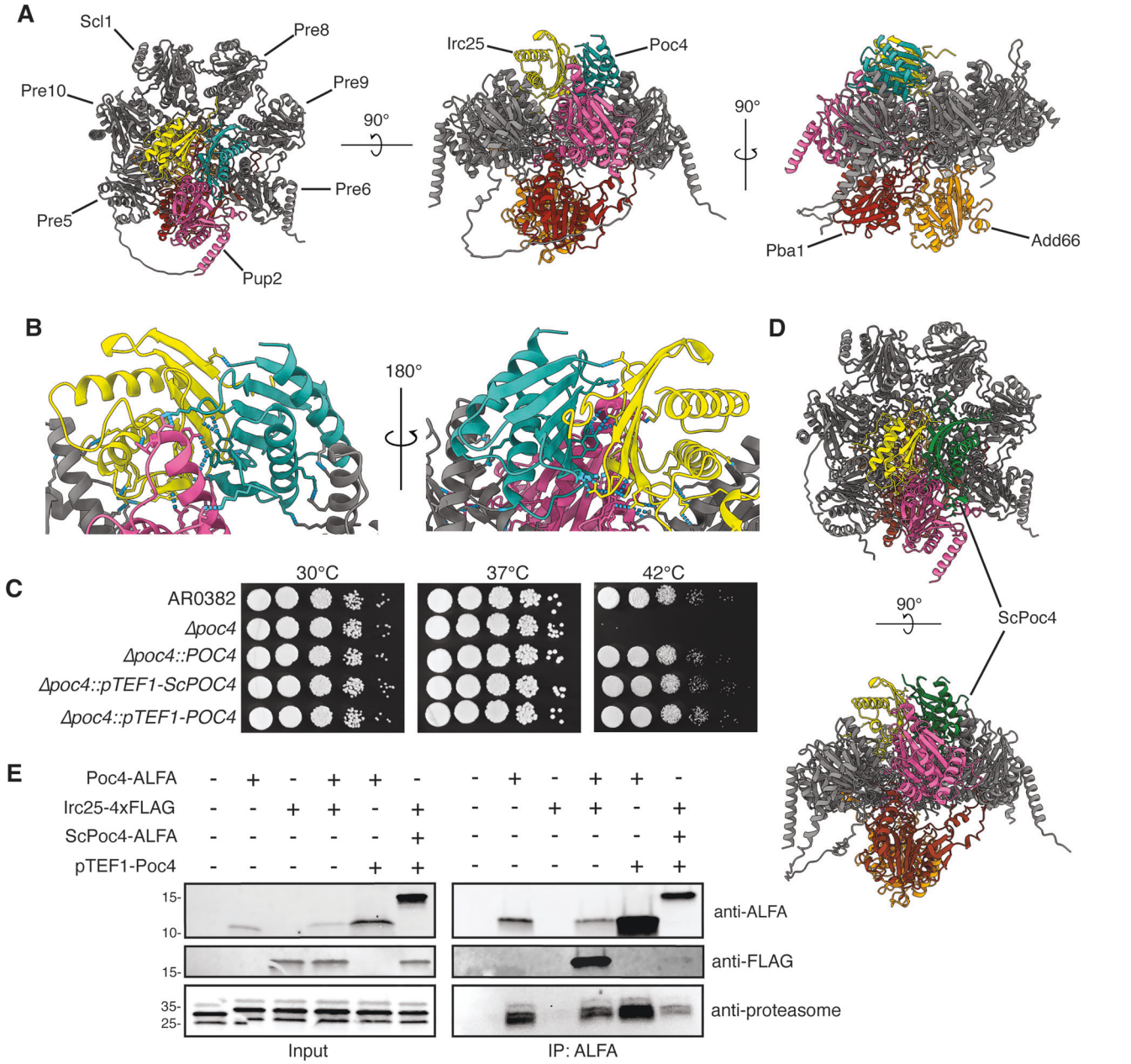


Fig. 4 | *S. cerevisiae* Poc4 can function in *C. auris*, despite sequence divergence. **A** AlphaFold3 model of *C. auris* proteasome subunits during core particle assembly. **B** Close-up view of Poc4-Irc25-Pup2 interface from structure in (**A**). Predicted hydrogen bonds indicated by dashed blue lines. **C** *S. cerevisiae* Poc4 is sufficient for growth at 42 °C in *C. auris*. Representative spot plates of indicated *C. auris* strains grown at 30, 37, and 42 °C. Indicated strains were diluted 10-fold, plated onto YPD agar, and incubated for 48 h before imaging. **D** AlphaFold3 model of *C. auris* proteasome, with *S. cerevisiae* Poc4 replacing *C. auris* Poc4 **E** *S. cerevisiae* Poc4 physically interacts with the 20S proteasome and Irc25. Co-immunoprecipitation of Irc25 and proteasome α-subunits with *C. auris* and *S. cerevisiae* Poc4 constructs. Numbers on the left indicate approximate size in kDa. Images and blots are representative of at least three biological replicates.

S. cerevisiae Poc4 can function in *C. auris*, despite considerable sequence divergence

The *C. auris* Poc4 role in proteasome function, despite its highly divergent sequence, suggests considerable flexibility in sequence-structure relationships within which evolution can operate while preserving essential functions. Recent advances in the prediction of multi-subunit protein complexes have made it possible to assess whether the physical interactions are preserved. We used AlphaFold3 to fold the proteasome core particle during the end of the alpha ring assembly, with the Pba1-Add66 and Irc25-Poc4 chaperone complexes on opposite faces of the proteasome, as this is thought to be the final step before Irc25-Poc4 dissociates, and thus this state represents the maximum point of potential protein-protein interactions for the chaperone complex. Starting with the *S. cerevisiae* complex as a baseline, we observed high predicted structural quality and similarity with the subset of the larger proteasome complex experimentally characterized by cryo-electron microscopy[31–33] (Supplementary Fig. 10, ipTM = 0.78, RMSD = 1.159 Å to 9D0T) and the Irc25-Poc4-Pup2 complex (RMSD = 0.646 Å to PDB: 2Z5C). This helped build confidence that the computationally predicted structure recapitulates native structures, and thus the quality of the prediction for the *C. auris* sequences. Using the same parameters, we folded the *C. auris* proteasome complex and also observed high predicted structural quality and structural similarity with the *S. cerevisiae* proteasome complex (Fig. 4A, Supplementary Fig. 11A, ipTM = 0.81). Furthermore, we observe high structural similarity between our predictions of the *C. auris* Irc25-Poc4-Pup2

complex and experimental structures from *S. cerevisiae* and *H. sapiens*, giving us confidence in our ability to model these higher order interactions (Supplementary Fig. 12). Although we observe discrepancies between the AF3 and experimental structures regarding loops and disordered regions, none of these regions are in the Pup2-Poc4-Irc25 subcomplex. Comparing the *S. cerevisiae* and *C. auris* complexes, we observe a conserved position of Poc4 around the pore of the proteasome with Pba1-Add66 on the opposite face. Notably, in addition to the conserved hydrogen bonds between the Irc25 and Poc4 protein along the dimerization interface, we observe interactions between helix-2 of Poc4 and helix-2 of Pup2, with polar interactions at the N-termini, hydrophobic packing along each helix, and the C-termini hydrophobically pack against each other, with the distance between the N-termini of each helix, (defined by the C-alpha atom of residue 83 of Pup2, and the C-alpha of residue 63 of Poc4) being 18.1 Å for *S. cerevisiae* and 19.3 Å for *C. auris*, suggesting that Poc4-proteasome interactions are crucial for chaperone activity (Fig. 4B, Supplementary Fig. 13).

Given these structural similarities despite only sharing only 20% sequence identity, we next sought to characterize whether the Poc4 homologs were functionally interchangeable between species. To understand the potential mechanism of rescue, we predicted the fold of the *C. auris* proteasome complex as above but replaced the endogenous Poc4 with *S. cerevisiae* Poc4 using AlphaFold3 (Fig. 4D, Supplementary Fig. 7E). Despite considerable sequence divergence, *S. cerevisiae* Poc4 is still predicted to interact with *C. auris* Irc25 and the Pup2 subunit with pDockQ2 scores of 0.42 and 0.25, respectively. The pDockQ2 score summarizes the predicted alignment error from AlphaFold3 over the interface residues within 15 Å between proteins, with scores above 0.2 corresponding to "Acceptable" docking as defined by Critical Assessment of Predicted Interactions community[34,35]. This suggested that although specific interaction residues were disrupted, the overall interaction interface is maintained. So, we tested if a CTG-optimized Poc4 allele from *S. cerevisiae* would complement the temperature-sensitive growth defects of the *C. auris* deletion strain (Fig. 4C). We expressed *ScPoc4* under a constitutive promoter in the *C. auris Δpoc4* background and observed full restoration of thermotolerance, suggesting that proteasome assembly activity is being rescued. Complementation of *Δpoc4* with a promoter-matched *C. auris* Poc4 also restores growth at 42 °C, but growth was not enhanced over that of the respective wildtype. Of note, we were unable to complement with the *S. cerevisiae* Poc4 allele under control of the endogenous *C. auris* Poc4 promoter because this strain failed to express (Supplementary Fig. 14). We then immunoprecipitated the Poc4 and found that in *C. auris, S. cerevisiae* Poc4 physically interacts with the proteasome α-subunits and Irc25 (Fig. 4E, Supplementary Fig. 4C, D), albeit to a lesser degree. Thus, we show a deeply conserved role of Poc4 across fungi that has persisted despite erosion of sequence signatures of homology.

Prospective testing of designed chaperone proteins

Our results suggest that certain structural features are sufficient for Poc4 proteasome assembly functions, but the generality of this inference is limited by the fact that Poc4 variants from the two species have a shared evolutionary history and have been shaped by natural selection. Meaning, despite the sequence divergence, the Poc4 protein in *S. cerevisiae* has also been selected for function as an assembly chaperone, and this selection bias could explain the functional rescue independent from the structural determinants of function. We hypothesized that prospectively testing the function of protein sequences outside of the evolutionarily accessible sequence space would allow us to understand the structural requirements for Poc4 activity. Towards this goal, we tested whether artificially designed sequences conditional on the structure of the native *C. auris* Poc4 expressed in *C. auris* would be sufficient to rescue the mutant phenotype (Fig. 5A), as these would not share the evolutionary bias of other naturally occurring Poc4 sequences. Given the higher conservation of the 29 residues at the Poc4 and Irc25 dimerization interface, we first predicted the Poc4 and Irc25 dimer using AlphaFold3 and then designed the remaining 82 positions of Poc4, considering any residue type except cysteine. To do this, we used Frame2Seq, a state-of-the-art masked language model for sequence design[36], which only considers the predicted backbone geometry of Poc4, and the predicted interaction with Irc25. We wanted to achieve a level of sequence diversity comparable to natural variation, so we used the default temperature parameter of 1.0. We generated 5000 designs, refolded each using AlphaFold2 and observed that for the 82 variable residues, recovery of the native Poc4 sequence ranged between 50 and 70% identity and 98.9% had an average pLDDT score >90, suggesting the scaffold is designable by Frame2Seq (Supplementary Fig. 15A). While this level of divergence is less than the 20% sequence conservation for *S. cerevisiae* to *C. auris*, to sample these sequences through unbiased deep mutational scanning would require testing >25^19 = 3.6e25 sequences.

To assess how the design sequences relate to naturally occurring Poc4 sequences, we leveraged embedding sequences into the continuous space learned by the ESM2 protein language model. In addition to the native *C. auris*, Poc4 and Irc25 sequences, and the redesigned Poc4, we also embedded the orthologous sequences identified through Foldseek and the POC3_POC4 pfam family (PF10448) and used UMAP to project the embedded representations to 2-dimensions. We observed clustering among the Poc4 homologs and among the Irc25 homologs. However, the designed structures appeared to be a distinct cluster (Supplementary Fig. 15B), which gave us confidence that our designs were distinct from natural Poc4 chaperones.

From these, we sampled 77 sequences for closer inspection, choosing 9 designs for experimental testing, with 61–64% sequence identity to *C. auris* Poc4 but without homology by BlastP to other proteins (Fig. 5B). We found that 7 of the designed proteins were able to be expressed in the *Δpoc4* deletion background, with 3 of the 7 expressed designs showing significant rescue of thermotolerance at 42 °C and one with partial rescue (Fig. 5C–E, Supplementary Fig. 16A, B). This data is further supported by immunoprecipitation, where we observed stronger associations with both Irc25 and the proteasome α-subunits with the functional Seq2523 design than the partially functional Seq2212 design (Supplementary Fig. 16C).

That some, but not all, designs were able to rescue *Δpoc4* suggested the structural features of the Poc4 fold substantially contribute to but do not fully explain the functional constraints on Poc4. During the design, we selected for interaction with Irc25 but not the rest of the proteasome. We hypothesized that examining Poc4 in the context of the remainder of the 20S proteasome would identify structural determinants that would allow for Poc4 function. Due to computational constraints, we used AlphaFold Server to predict the structure of 466 designs in complex with the 20S proteasome subunits, and then computed the max pDockQ2 score between Poc4 and Pup2. We observed that the median score was 0.05 and only 41 (8.7%) had scores above the previously published[34] threshold of 0.2 (Fig. 6A). To enrich for designs with better pDockQ2 scores, we trained a regression model to predict pDockQ2 based on ESM2 embeddings of the Poc4 designs using H2O AutoML. Using this ML predictor, we selected an additional 136 top-scoring designs to prospectively fold with the AlphaFold Server and observed an improved median pDockQ2 of 0.13, with 33 designs giving a pDockQ2 > 0.2 (Fig. 6A, Supplementary Fig. 17A, AUROC = 0.73).

To identify the most important interactions underlying the pDockQ2 scores, we first computed PLiP interaction features[37], summarized as the counts across all Pup2 residues by the 8 interaction types, yielding 77 features. We then trained a regression-ML model to

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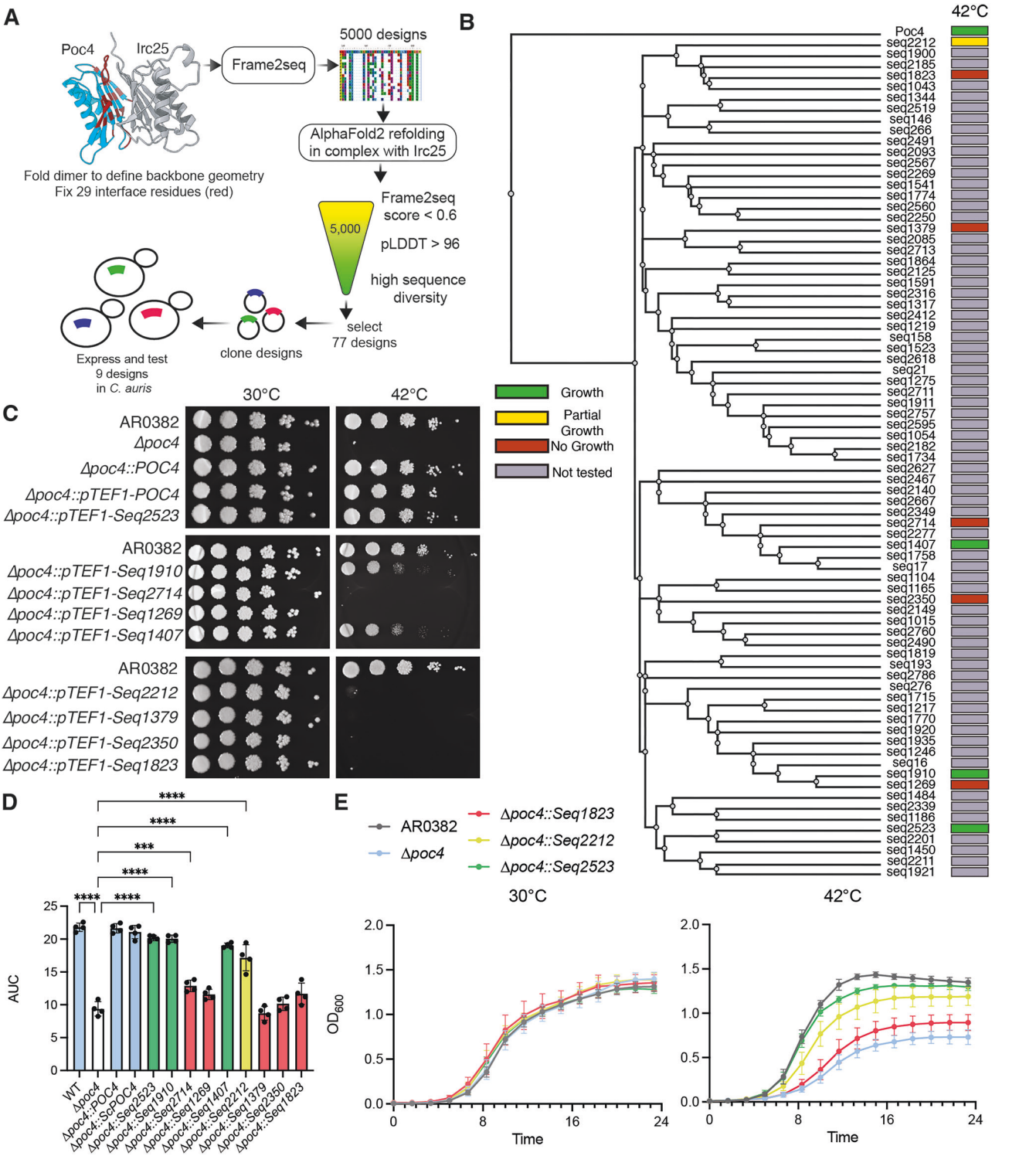


Fig. 5 | Poc4 chaperone designs are functional in vivo. A Schematic of the Poc4 re-design process. Briefly, the Frame2Seq model was generated based on the Poc4-Irc25 backbone to generate 5,000 sequences. Ultimately, 9 of these designs were tested in *C. auris* by cloning a codon-optimized version into the *POC4* locus under the *pTEF1* promoter and assessing for growth at 42 °C. **B** Phylogram of 77 design sequences chosen for manual inspection. Bars are colored based on the ability of *Δpoc4 C. auris* strain to grow at 42 °C when complemented with the designed sequences. The native Poc4 protein sequence serves as the outgroup. **C** Chaperone designs rescue *C. auris* growth at 42 °C. Representative spot plates of the indicated *C. auris* strains grown at 30 and 42 °C. Indicated strains were diluted 10-fold, plated onto YPD agar, and incubated for 48 h before imaging. **D** Quantification of indicated *C. auris* strains' growth at 42 °C by area under the curve (AUC). Statistical significance among different groups was calculated using one-way ANOVA, with Tukey's post hoc tests for multiple comparisons, and *p*-values for all comparisons are listed in Source Data. ***$p = 0.001$, ****$p < 0.0001$. Chaperone designs with no symbol had $p > 0.05$. **E** Representative growth curves of indicated *C. auris* strains at 30 and 42 °C. Growth was measured by OD_{600} every 10 min for 24 h. Data shown is the mean ± standard deviation across 4 biological replicates every 80 min.

https://doi.org/10.1038/s41467-026-71206-4

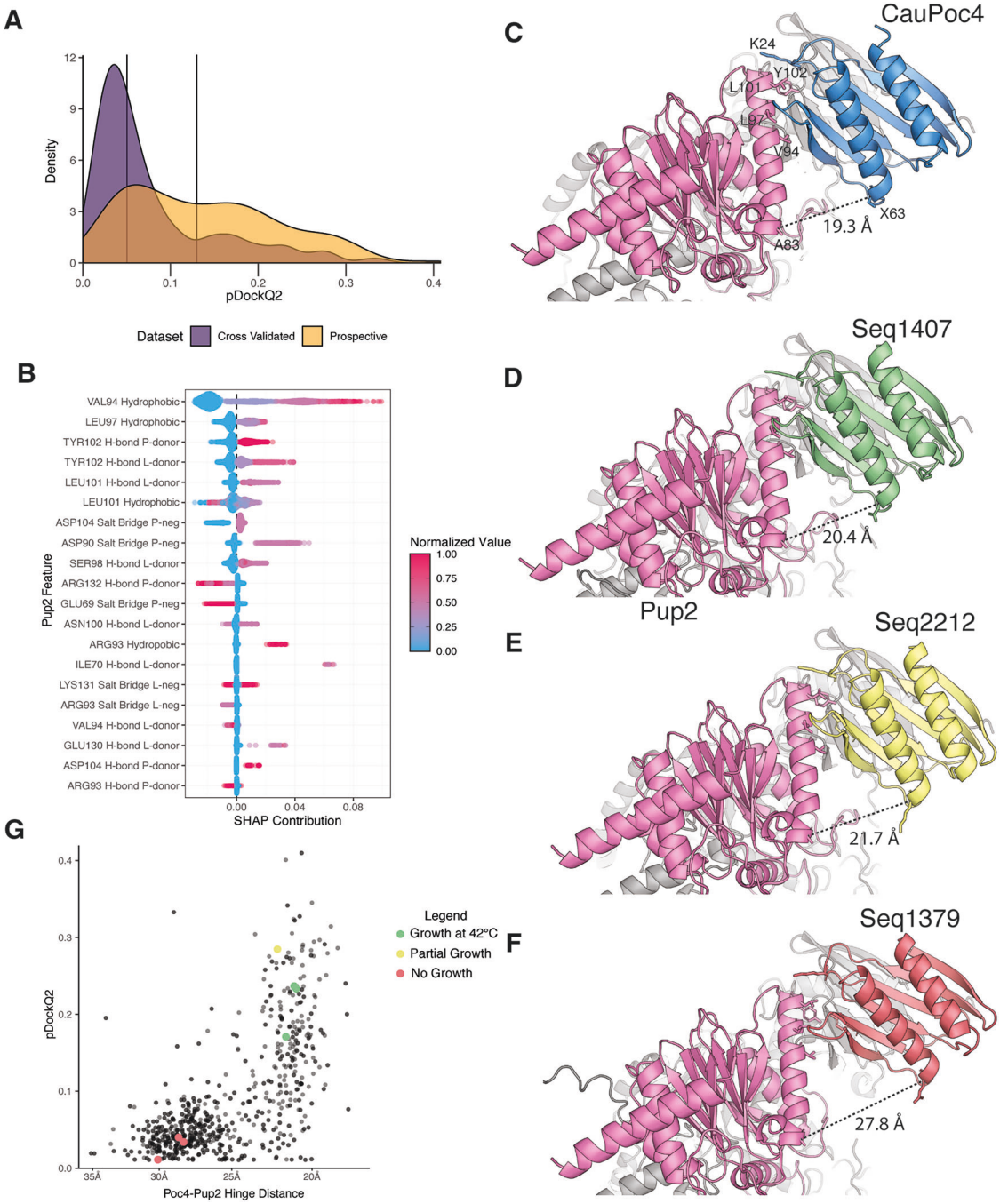


Fig. 6 | Identifying features of designed proteins that improve docking. **A** Kernal density estimation of pDockQ2 scores of the 466 cross-validated and 136 prospectively identified structures. Vertical lines indicate the median for each set. **B** Top SHAP features for Pup2-Poc4 interactions. **C**–**F** AlphaFold3 model of *C. auris* proteasome subunits during core particle assembly, with designed chaperone instead of the native Poc4. Dashed line indicates the measured distance between A83 of Pup2 and Res63 on Poc4. Pup2 is shaded pink, while Poc4 chaperones are colored based on activity, with the native sequence in blue. **G** Plot of the Poc4-Pup2 hinge distance compared with pDockQ2 scores demonstrates a negative correlation ($\rho = -0.65$). Experimentally tested designs are marked and colored by functionality.

predict the pDockQ2 scores from the interaction features and estimated the top SHAP features. The top features were hydrophobic interactions with Pup2 residues 94, 97, and 101, followed by polar interactions with residues 101 and 102 (Fig. 6B, Supplementary Fig. 17B, C). Inspecting these features for the wildtype *C. auris* and *S. cerevisiae* complexes and the 7 tested designs, we found that for the wildtype protein and designs that fully or partially recover function, sheets 5 and 6 of Poc4 (Res41-Res58) pack against helix-2 of Pup2 (Res83-Res102), thus allowing K24 and Y44 in WT-Poc4 and the Poc4 designs to form H-bonds with the C-terminal backbone carbonyl of helix-2 of Pup2 (Y102 and D104, respectively) (Fig. 6C–F, Supplementary Fig. 18). To quantify this observation, we defined the Pup2-Poc4 packing distance to be the distance between the C-terminal of helix-2 Pup2 (A83, Cα) and the C-terminal of helix-2 of Poc4 (Res63/Cα). Plotting the pDockQ2 against packing distance, we see a spearman rank correlation of -.65 (Fig. 6G), suggesting that tight packing is a requirement for good pDockQ2 scores. Assessing the structural features of the 7 experimentally tested designs, we can discriminate between non-functional designs and those that partially or fully recover activity. Notably, of the designs that fully or partially recover activity, the

design with partial recovery of activity has a higher score than those that fully recover activity, consistent with pDockQ2 being a good but incomplete predictor of function.

Discussion

Over the past decade, advances in genome sequencing have drastically increased the number of available gene/protein sequences, and our ability to compare and define relationships between sequences is critical for our understanding of genome function, evolution, and biotechnology. Central to this process are sequence-based search algorithms, such as BLAST, which have revolutionized our understanding of comparative genomics and evolution. However, these methods often fail to identify true homologs once sequence identity falls below 30%, reaching the so-called twilight zone[1]. This issue has widespread effects: recent estimates suggest that 40–45% of all lineage-restricted genes in fungi are due to homology detection failure and not de novo origination[38,39]. Consequently, homologous proteins that evolve rapidly in sequence have been generally excluded from comparative studies on protein evolution. Here, we leveraged comparisons of protein structure to overcome the limitations of sequence analysis and identify cases of structural homology in the emergent fungal pathogen *C. auris*. This work demonstrates how we can use structure to identify proteins within the twilight zone, as well as improve hypotheses about salient structural features and test those hypotheses at the biochemical and cellular levels.

When focusing on proteins within the BlastP twilight zone, we identified a hypothetical protein with structural similarity to the Poc4 assembly chaperone in *S. cerevisiae*. We found that yeast proteasome assembly chaperones rapidly evolve at rates that would obfuscate homology detection by sequence-based approaches. Yet, despite the sequence divergence between *C. auris* Poc4 and *S. cerevisiae* Poc4, we found that the orthologs have evolutionarily conserved function in proteasome assembly and are functionally interchangeable in *C. auris*. This was especially important to test experimentally, as many previous studies on proteins in the twilight zone have only identified the relationship by structural comparisons and have not tested whether the proteins are functional across systems. For example, recent work from Riziotis and colleagues demonstrated that enzymes with highly diverged functions still maintain regions of conserved active site structures[29]. Findings from ancestral sequence reconstruction studies also highlight why structural conservation alone is not sufficient for inference of conserved function: a consistent theme from the study of historical protein evolution using this method is that evolutionary changes in protein specificity and function can be recapitulated with fewer than three mutations at contact sites, while the rest of the structure remains conserved[40,41]. Observations from directed evolution studies complement these findings from studies of natural evolution—selection for efficient nicotine degradation in the NicA2 enzyme led to >100× increase in the catalytic rate of the enzyme while the crystal structures of the pre- and post-evolved variants remained virtually identical (RMSD < 0.5 Å)[42]. However, the fact that ASR could be performed on those proteins means that the rate of protein sequence evolution was sufficiently slow that amino acid states at most sites could be confidently inferred. The preservation of sequence signal in the evolutionary interval of interest is necessary for ASR and is a reason that certain types/domains of proteins with rapidly evolving sequences or unknown structures, such as IDRs, have not been investigated. Additionally, the prevalence of protein moonlighting and neofunctionalization suggests that even a conserved structure may not be sufficient to predict function, since potentially different biochemical capacities of a protein can be revealed in different cellular contexts[29,43]. Collectively, these studies emphasize the need for direct experimental determination of conserved function following hypotheses generated from bioinformatic analyses.

Our initial observation that proteasome assembly chaperones are rapidly evolving was surprising because the client proteins of these chaperones, the subunits of the proteasome, are highly conserved in sequence and structure throughout Eukarya[44,45]. The proteasome itself is responsible for the majority of regulated protein degradation and plays major roles in cellular function. Furthermore, specialized forms of the proteasome exist in higher eukaryotes, playing key roles in the immune response, and proteasome dysfunction is linked to cancer and neurodegeneration in humans. However, the precise role of these alternative proteasome isoforms, as well as their assembly and regulation, is not well understood. While the sequence divergence and functional redundancy between human and yeast proteasome assembly chaperones were noted during their initial discovery, the rapid divergence and evolutionary implications were not explored[23,24]. We had also expected that proteins that physically interact and codetermine function would also have similar coevolutionary rates[6]. However, our observations suggest this does not have to be the case; rather, they support the idea that in a given system, a wide range of tolerable sequences may be sufficient for a specific structure that enables a specific function[46]. In the absence of a change in structure, there is then nothing for selection to act upon, thus resulting in the accumulation of sequence changes over time. This may be additionally exacerbated in the CTG clade, as many of these organisms do not regularly mate and allow for gene flow and recombination; however, the long branch lengths in close relatives to *S. cerevisiae*, which do undergo mating and recombination, suggest it is not the only factor[47]. Furthermore, as the Poc4 sequence diverges in a lineage, epistatic interactions between and within chaperone proteins and proteasome subunits may further entrench sequence changes, preventing any sort of reversion back to an ancestral sequence[16,48–50].

Protein evolution is strongly shaped by historical contingency, meaning the paths available to and taken by evolution are strongly affected by prior evolutionary events, with selection potentially acting at each historical intermediate[49,50]. A consequence of this contingency is that many parts of protein sequence space likely exist that have not been sampled over evolution. To explore this unsampled or otherwise evolutionarily inaccessible protein space, we leverage recent advances in protein design and prospective testing to evaluate the sufficiency of a specific fold to fulfill a function required for cellular survival. The scale and feasibility of protein design have greatly increased with our ability to predict protein structures from a sequence, synthesize them, and validate their functions[9,10,51,52]. We and others have previously published examples of redesigned enzymes with higher efficiency or stability than their natural counterparts, with a range of medical and technical applications[53–56]. Nevertheless, most designed proteins are generated for in vitro use, and few examples exist of such proteins being functional in vivo[57,58]. In our approach, we were able to simultaneously mutate 40 of the 111 residues of Poc4 and still maintain function. Through total protein re-design, it is possible to explore epistatic relationships between amino acids and potentially access high fitness variants that are unlikely to emerge through stepwise evolutionary approaches. This approach expands the sequence space of functional proteins beyond what can be achieved through deep mutational scanning or natural and lab evolution of new traits and functions, as these single-step approaches can be limited by constraints imposed by epistatic interactions within the protein[15,59,60].

A limitation of this work is that our designed sequences maintained some identity with the natural variant, instead of a 0% identity replacement. Specifically, we fixed 29 of the 111 Poc4 residues that were along the Poc4-Irc25 interface, and our final designs have 60–65% identity to the native Poc4 protein. Further work with sequence design with different patterns of fixed residues may lead to lower identity between designs and natural sequences. Additionally, our ability to generate high-confidence AlphaFold models of the *C. auris* Poc4 structure is due to experimentally determined structures of divergent homologs with conserved folds present in the AlphaFold training data. This bias may limit the feasibility of our approach for

other protein families where a representative crystal structure has not yet been solved.

We also determined that not all of the designed artificial chaperones were functional, suggesting additional constraints that were not captured in our original approach. Additional experiments regarding the biochemical properties, such as stability or aggregation potential, of the designed proteins may help explain the different functional capacity of our designs. We find that computational measures of protein docking are sensitive enough to identify potential differences between functional and non-functional designs. Wildtype and designs that fully or partially recover function, form a hydrophobic pack against Pup2. Interestingly, the two designs that formed the highest quality interactions with Pup2 only partially recovered function, suggesting perhaps that moderately strong interactions facilitate transient chaperoning of functional roles. This observation, along with other studies that directly integrate diffusion and docking models into protein design workflows[54,61], suggests that modeling endogenous protein-protein interactions of designed sequences is a critical step in identifying protein designs with desired activity[54]. Combined with high-throughput testing of designs in a system amenable to high-throughput genetics, such as *S. cerevisiae*, this "design and test" approach may provide a powerful tool to comprehensively map the functional capacity of protein sequence space. Similar experiments have shown the functional capacity of evolutionarily inaccessible sequences in the context of a protein through combinatorial mutagenesis; however, these have been limited to only a small region of a protein[62].

A strength of the study is that we did not require an experimentally characterized *C. auris* protein structure to perform the design. However, we recognize that our ability to design and test these chaperones is greatly facilitated by the small, single-domain nature of the target protein and a simple phenotypic test for function. As proteins can vary greatly in size, domain, architecture, and function, this approach will not be feasible for all proteins. It is also possible that redesign is only possible for proteins that already show high rates of sequence divergence between organisms—there may be higher degeneracy between sequence and structure in such proteins, making it possible for many alternative sequences to encode the same structure without incurring fitness costs. Overall, our work provides an example where we can modulate each aspect of the sequence-structure-function relationship to understand the functional constraints on protein evolution.

Methods

Strains and culture conditions

A list of all strains used in this study is included in Supplementary Data 1. Clinical *C. auris* isolates were obtained through the CDC Antibiotic Resistance Isolate Bank. Unless specified, cells were cultured at 30 °C in YPD liquid media (1% yeast extract, 2% peptone, 2% dextrose), with constant shaking. For a solid medium, 2% agar was used. All strains were maintained as frozen stocks in 25% glycerol at −80 °C.

Genomic DNA extraction

Genomic DNA was isolated using a PCA extraction method. Yeast cells were cultured overnight in YPD liquid at 30 °C and harvested by centrifugation. Cells were resuspended in lysis buffer (2% (v/v) Triton X-100, 1% (w/v) SDS, 100 mM NaCl, 10 mM Tris-Cl, 1 mM EDTA) and disrupted by bead beating. Released DNA was extracted into PCA and then into chloroform. The resulting DNA was then purified by ethanol precipitation and resuspended in water before treatment with RNase A. RNase was heat-inactivated, and the extracted DNA was further purified by ethanol precipitation and resuspended in water.

Primers and plasmids

A list of all plasmids used in this study is included in Supplementary Data 2. A list of all primers used in this study is included in Supplementary Data 3. Cassettes for transformation of *C. auris* were maintained in the multiple cloning site of the pUC19 cloning vector and assembled from fragments using the NeBuilder HIFI DNA Assembly Master Mix (NEB #E2621) or the Codex DNA Gibson Assembly Ultra Master Mix (Codex DNA #GA1200) according to the manufacturers' instructions. All plasmid assemblies were confirmed by Nanopore sequencing.

Plasmid and strain construction

pTO301/CauTO467: The plasmid backbone was amplified from pTO139 using oTO590 and oTO591. The NAT cassette was amplified from pTO137 using oTO847 and oTO668. The NAT cassette was flanked by 500 bp of regions homologous to the regions immediately up and downstream of *POC4* (B9J08_000884) amplified from *C. auris* AR0382 genomic DNA using oTO1734-oTO1735 and oTO1736-oTO1737, respectively. The repair cassette was amplified using oTO18 and oTO19 and transformed into AR0382 to generate CauTO467 and CauTO468 from independent transformations.

pTO306/CauTO497: The plasmid backbone was amplified from pTO139 using oTO590 and oTO591. The ADH1 terminator and NEO cassette were amplified from pTO169 using oTO1066 and oTO668. The 500 bp upstream and ORF of *POC4* (B9J08_000884) was amplified from *C. auris* genomic DNA using oTO1734 and oTO1842, while the downstream 500 bp of homologous sequence was amplified from oTO1736 and oTO1737. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO497.

pTO315/CauTO496: The plasmid backbone was amplified from pTO139 using oTO590 and oTO591. The NAT cassette was amplified from pTO137 using oTO847 and oTO668. The NAT cassette was flanked by 500 bp of regions homologous to the regions immediately up and downstream of *IRC25* (B9J08_003917) amplified from *C. auris* genomic DNA using oTO1928-oTO1929 and oTO1930-oTO1931, respectively. The repair cassette was amplified using oTO18 and oTO19 and transformed into AR0382 to generate CauTO496.

pTO322/CauTO550: The plasmid backbone was amplified from pTO139 using oTO590 and oTO591. The ADH1 terminator and NEO cassette were amplified from pTO169 using oTO1066 and oTO668. The NEO cassette was flanked upstream by the ORF of *IRC25* (B9J08_003917) and 500 bp of upstream sequence, which was amplified from *C. auris* genomic DNA using oTO1928 and oTO1980. A downstream flanking sequence was amplified from *C. auris* genomic DNA using oTO1930-oTO1931. The repair cassette was amplified using oTO18-oTO19 and was transformed into CauTO496 to generate CauTO550.

pTO324/CauTO546: The plasmid backbone was amplified from pTO139 using oTO590 and oTO591. The NAT cassette was amplified from pTO137 using oTO874 and oTO668. The NAT cassette was flanked by 500 bp of regions homologous to the regions immediately upstream and downstream of *RPN4* (B9J08_003372) amplified from *C. auris* genomic DNA using oTO1983-oTO1984 and oTO1985-oTO1986, respectively. The repair cassette was amplified using oTO18 and oTO19 and transformed into AR0382 to generate CauTO546.

pTO346/CauTO604: The plasmid backbone was amplified from pTO139 using oTO590 and oTO591. The NAT cassette was amplified from pTO137 using oTO874 and oTO668. The NAT cassette was flanked by 500 bp of regions homologous to the regions immediately upstream and downstream of *PRE9* (B9J08_002268) amplified from *C. auris* genomic DNA using oTO2125-oTO2126 and oTO2133-oTO2134, respectively. The repair cassette was amplified using oTO18 and oTO19 and transformed into AR0382 to generate CauTO604.

pTO370/CauTO629: The plasmid backbone was amplified from pTO139 using oTO590 and oTO591. The ADH1 terminator and NEO cassette were amplified from pTO169 using oTO1066 and oTO668. The NEO cassette was flanked upstream by the ORF of *PRE9* (B9J08_002268) and 500 bp of upstream sequence, which was amplified from *C. auris* genomic DNA using oTO2125 and oTO2210. A

downstream homologous flanking sequence was amplified from *C. auris* genomic DNA using oTO2133 and oTO2134. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO604 to generate CauTO629.

pTO325/CauTO552/CauTO943: The entire pTO306 vector was amplified in two overlapping fragments using overlapping backbone-specific primers oTO1427 and oTO1428 paired with oTO1958 and oTO1959, respectively. oTO1958 and oTO1959 contain the C-terminus of POC4 with an ALFA epitope tag directly before the stop codon. The repair cassette was amplified using oTO18-oTO19 and transformed into CauTO467 and CauTO496 to generate CauTO552 and CauTO943, respectively.

pTO341/CauTO591: The *POC4* complementation vector was amplified from pTO306 using oTO2106 and oTO1066. A CUG-optimized and C-terminal ALFA-tagged *POC4* gene based off *S. cerevisiae* S288C *POC4* was obtained as a gene fragment from Twist Biosciences (San Francisco, California, USA) and assembled with the amplified fragment to generate pTO341. The repair cassette was amplified using oTO18-oTO19 and transformed into CauTO467 to generate CauTO591.

pTO350/CauTO611: 1000 bp of promoter sequence upstream of *TEF1* (B9J08_003610) was amplified from *C. auris* genomic DNA using oTO1168 and oTO1169. The entire pTO325 cassette was amplified with pTO2178 and oTO2179 and assembled to incorporate p*TEF1* directly upstream of the *ScPOC4* ORF. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO611.

pTO352/CauTO612: 1000 bp of promoter sequence upstream of *TEF1* (B9J08_003610) was amplified from *C. auris* genomic DNA using oTO1168 and oTO1169. The entire pTO325 cassette was amplified with oTO2179 and oTO2184 and assembled to incorporate p*TEF1* directly upstream of the *POC4* ORF. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO612.

pTO381/CauTO689: The IRC25 deletion cassette was amplified from pTO315 using oTO1427 and oTO2343 and combined with the NEO cassette from pTO322 amplified using oTO669 and oTO1428. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO689.

pTO435/CauTO804/CauTO805/CauTO806/CauTO807/CauTO808/CauTO942: The IRC25 complementation cassette was amplified into three fragments to incorporate a C-terminal 4xFLAG construct. The first fragment was amplified from pTO322 using oTO2717 and oTO2749, while the second fragment was amplified using oTO2748 and oTO2718. The HYG resistance cassette was amplified using oTO2719 and oTO2720. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO496 to generate CauTO804. The repair cassette was transformed into CauTO552, CauTO611, CauTO799, CauTO800, and CauTO467 to generate CauTO805, CauTO806, CauTO807, CauTO808, and CauTO942 respectively.

pTO421/CauTO799: The *POC4* high-expression complementation vector was amplified from pTO350 using oTO1169 and oTO1066. The Seq2523 protein was synthesized as a CUG-optimized gene fragment containing a C-terminal ALFA tag from Twist Biosciences and assembled with the complementation vector to generate pTO421. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO799.

pTO422/CauTO800: The *POC4* high-expression complementation vector was amplified from pTO350 using oTO1169 and oTO1066. The Seq2212 protein was synthesized as a CUG-optimized gene fragment containing a C-terminal Alfa tag from Twist Biosciences and assembled with the complementation vector to generate pTO422. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO800.

pTO423/CauTO801: The *POC4* high-expression complementation vector was amplified from pTO350 using oTO1169 and oTO1066. The Seq1379 protein was synthesized as a CUG-optimized gene fragment containing a C-terminal Alpha tag from Twist Biosciences and assembled with the complementation vector to generate pTO423. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO801.

pTO424/CauTO802: The *POC4* high-expression complementation vector was amplified from pTO350 using oTO1169 and oTO1066. The Seq2350 protein was synthesized as a CUG-optimized gene fragment containing a C-terminal Alfa tag from Twist Biosciences and assembled with the complementation vector to generate pTO424. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO802.

pTO425/CauTO803: The *POC4* high-expression complementation vector was amplified from pTO350 using oTO1169 and oTO1066. The Seq1823 protein was synthesized as a CUG-optimized gene fragment containing a C-terminal Alpha tag from Twist Biosciences and assembled with the complementation vector to generate pTO425. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO803.

pTO463/CauTO957: The *POC4* high-expression complementation vector was amplified from pTO350 using oTO1169 and oTO1066. The Seq1910 protein was synthesized as a CUG-optimized gene fragment containing a C-terminal ALFA tag from Twist Biosciences and assembled with the complementation vector to generate pTO447. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO957.

pTO464/CauTO958: The *POC4* high-expression complementation vector was amplified from pTO350 using oTO1169 and oTO1066. The Seq2714 protein was synthesized as a CUG-optimized gene fragment containing a C-terminal ALFA tag from Twist Biosciences and assembled with the complementation vector to generate pTO447. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO958.

pTO465/CauTO959: The *POC4* high-expression complementation vector was amplified from pTO350 using oTO1169 and oTO1066. The Seq1269 protein was synthesized as a CUG-optimized gene fragment containing a C-terminal ALFA tag from Twist Biosciences and assembled with the complementation vector to generate pTO447. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO959.

pTO467/CauTO960: The *POC4* high-expression complementation vector was amplified from pTO350 using oTO1169 and oTO1066. The Seq1407 protein was synthesized as a CUG-optimized gene fragment containing a C-terminal ALFA tag from Twist Biosciences and assembled with the complementation vector to generate pTO447. The repair cassette was amplified using oTO18 and oTO19 and transformed into CauTO467 to generate CauTO960.

C. auris Transformation

C. auris transformation was performed using a transient-Cas9 expression system as described previously[63]. Transformation repair cassettes were amplified from assembled plasmids. The Cas9 expression cassette was amplified from pTO135 with oTO40 and oTO41. Cassettes for the expression of sgRNA targeting specific loci were amplified from pTO136 using overlap extension PCR to change the gRNA sequence. All linear PCR products were purified using a Zymo DNA Clean & Concentrator kit (Zymo Research, D4034) according to the manufacturer's instructions.

To prepare competent *C. auris* cells, a single colony was inoculated into liquid YPD and incubated overnight at 30 °C with gentle shaking. Cells were harvested by centrifugation and resuspended in TE Buffer and 100 mM lithium acetate. Cells were further incubated at 30 °C for 1 h with constant shaking before DTT was added to the cells at

a final concentration of 25 mM before further incubating at 30 °C for an additional 30 min. Cells were harvested by centrifugation at 4 °C before being washed once with ice-cold water and once with ice-cold 1 M Sorbitol. Harvested cells were resuspended in ice-cold 1 M Sorbitol and kept on ice for immediate use or stored at −80 °C for future use.

Electroporation was performed by adding 50 μL of competent cells to a pre-chilled electro-cuvette (2mm-gap) with 500 ng of each of the PCR-amplified Cas9, sgRNA, and repair cassettes. Cells were electroporated using a Bio-Rad MicroPulser Electroporator using the pre-defined *P. pastoris* (PIC) protocol (2.0 kV, 1 pulse). Electroporated cells were immediately recovered in 1 M sorbitol and then resuspended in YPD for a 2-h outgrowth at 30 °C with constant shaking. Outgrowth cells were plated onto selective media. For repair cassettes containing the *NAT* marker, cells were selected on YPD plates containing 200 μg/mL nourseothricin with incubation at 30 °C for 2 days. For repair cassettes containing the *NEO* marker, cells were selected on YPD plates containing 1 mg/mL G418 with incubation at 30 °C for 2 days. For repair cassettes containing the *HYG* marker, cells were selected on YPD plates containing 800 μg/mL Hygromycin B with incubation at 30 °C for 2 days.

Transformant colonies were passaged to isolation, and correct incorporation of the repair cassette was confirmed for each by colony PCR using Phire Plant Direct PCR Master Mix (F160; Thermo Fisher Scientific) according to the manufacturer's instructions. Each mutant was confirmed with at least three independent PCR reactions with distinct primer sets specific to the mutation site and compared to parental strains.

Phylogenetic analysis

The protein sequences used in the phylogenetic analysis are available in Supplementary Data 4. Sequence homologs of the indicated genes were identified using the *S. cerevisiae* amino acid sequence as the query using NCBI BLAST. For *CauPOC4* homologs, the *C. auris POC4* amino acid sequence was used instead. The sequence was individually queried against each of the listed species using the blastp algorithm in the non-redundant protein sequences (nr) database under default parameters. Hits were identified by having reciprocal best matches and an E-value less than 0.0001. Syntenic genes were identified in the listed species using both the Candida Gene Order Browser and the Yeast Gene Order Browser[64,65].

For each gene, the homologous sequences were first aligned using muscle5 and trimmed using trimAl (parameters: -gt 0.7 -cons 60) and then manually adjusted to remove unalignable sites from downstream analyses[66,67]. The curated sequence files used in downstream analyses are provided in Supplementary Data 4. These alignment files were used to infer the maximum likelihood phylogeny for each gene using the best-fit model of sequence evolution and rate variation parameters using the IQTree2 software and the implemented ModelFinder function[68,69]. This method provided a direct estimate of the evolutionary distance between pairs of sequences in the alignment.

Per-residue conservation scores were calculated from these alignments using the AL2CO method with the following settings: unweighted sequences, sum-of-pairs conservation method, BLOSUM62 scoring matrix without transformation, window size of 1, gap fraction of 0.5[70]. Scores were visualized onto the *C. auris* Poc4 and Irc25 AlphaFold3 models using ChimeraX.

Homology analysis

For sequence-based homology, 5,417 protein-coding sequences from FungiDB (version 68) CaurisB8441 were searched using BlastP (Protein-Protein BLAST 2.12.0+) against all 572,970 Swiss-Prot (version 2025_01). For structure-based homology, the same protein coding sequences were mapped to structures in the AFDB (v4, model creation date: 2022-06-01) as follows. First, the sequences were mapped to locus tags in the NCIB GCA_002759435.3_Cand_auris_B8441_V3 genome, yielding 5395 one-to-one matches. Note FungiDB CaurisB8441 ids have the form B9J08_###### (6 numbers), while GCA_002759435.3_Cand_auris_B8441_V3 ids have the form B9J08_##### (5 numbers). Then, GCA_002759435.3_Cand_auris_B8441_V3 locus tags were mapped UniPort entries for the UP000230249 Proteome, where all ids mapped 1-1. Each UniProt entry was then mapped to UniParc IDs, which uniquely identifies the sequence. Then, the AFDB was queried for structures for *C. auris*, yielding 12,017 structures, and their pLDDT scores were recorded. For each AFDB structure, the corresponding UniProt entry was mapped its UniParc entry and de-duplicated. Joining *C. auris* protein and AFDB structures by UniParc IDs yielded 4360 AFDB structures for *C. auris* proteins. We then used Foldseek (downloaded 2/2025, Foldseek and MMseq version 9-427df8a) to search for structure homologues in the pre-defined Alphafold/Swiss-Prot reference database, using the foldseek easy-search command with default parameters otherwise. To compare sequence vs structure homology, for each *C. auris* gene, we computed the maximum BlastP and Foldseek bit score across Swiss-Prot targets.

Structural predictions

Protein models of mid-assembly proteasome assembly complex were predicted using AlphaFold3 (https://alphafoldserver.com/ 5/2024) with default parameters. For each structure, the top-ranking model was visualized with ChimeraX, and predicted aligned error (PAE) plots were made with PAE Viewer[71,72]. Predicted intermolecular interactions between chains were identified using PISA[73]. pDockQ2 scores were calculated between all chains in a given structure using *the IPSAE package* as an average across 5 models[74]. Poc4 chaperone topology was determined using Pro-origami under default parameters[75].

Protein design

As an initial structure for Poc4, the *C. auris* Poc4 and Irc25 sequences were jointly folded with AlphFold3 (https://alphafoldserver.com/ 5/2024) with default parameters, and the top-ranking structure was used, having scores pTM: 0.89, ipTM: 0.91, has_clash: 0.0, fraction_disordered: 0.01, and ranking_socre: 0.91. Then given this structure, Frame2Seq (version 0.0.5) was used to generate designs for the Poc4 chain, generated excluding cysteine, varying temperature in [0.01, 0.1, 1, 10], and either using no additional constraints, fixing conserved residues determined by the multiple sequence alignment [18, 24, 26, 42, 44, 52, 57, 67, 76, 78, 79, 83, 85, 86, 89], or alternatively fixing the Poc4 residues at the Poc4-Irc25 interface [1, 3, 5, 12, 14, 16, 18, 24, 25, 26, 28, 30, 32, 33, 34, 35, 36, 37, 38, 39, 42, 44, 55, 56, 57, 58, 83, 85, 87]. From the temperature = 1.0 and fixinterface parameters, 5000 sequences were generated, and each was folded with AlphaFold2. ParaFold was used to distribute first the MSA generation and then the folding tasks onto the University of Michigan Great Lakes CPU and GPU HPC cluster queues[76]. Designs were evaluated by the Frame2Seq Score, AlphaFold2 pLDDT, sequence identity to the native Poc4, and diversity among the designs. To visualize the similarity between designs, Poc4 and structural homologues, ESM2 embeddings for the Poc4 Foldseek hits, members of pfam PF10448, and the designs were generated using the esm2_t33_650M_UR50D model, by averaging the per-residue representation in the 33rd layer for the up to 1022 first residues in the sequence. The EMS2 embeddings were then projected to 2-dimensions using UMAP (v0.5.7) with parameters $a = 1$, $b = 1.8$, metric = "cosine".

Structural features analysis

To characterize the structural basis for successful Poc4, we analyzed the pDockQ2 score between an additional 602 Poc4 designs and Pup2 in AlphaFold3 predicted proteasome complexes. To do this, we generated an additional 500,000 Frame2Seq sequences and filtered for Frame2Seq score <0.6, yielding 44,119 designs (8.8%). From these, we

randomly sampled 5141 sequences from the lowest 10,000 by percent sequence recovery (so that percent recovery ranged between 57 and 63%), used AlphaFold2 to re-fold the Poc4-Irc25 complex, filtered for pLDDT > 96, yielding 2268 designs (44%). From these, we randomly sampled 466 designs and co-folded each with the 10 other components of the *C. auris* proteasome complex on the AlphaFold Server and computed pDockQ2 scores for each of the 5 generated sampled conformations for each complex. To compute the pDockQ2 score of Pup2 and Poc4 in each complex, we used the IPSAE (v3 03/2025) package with default pae_cutoff=15 and dist_cutoff=15 parameters, and since pDockQ2 is an asymmetric measure, we used the max over Pup2 vs. Poc4 and Poc4 vs. Pup2. We observed that the median pDockQ2 score was of 0.05, and 41 had a pDockQ2 score >0.2. To balance the distribution of pDockQ2 scores for comparative analysis, we trained a regression model to predict pDockQ2 based on ESM2 embeddings of the Poc4 designs using H2O AutoML (v3.44.0.3). The best performing model over 5-fold cross-validation by RMSD was a Gradient Boosted Model, with an RMSD of 0.061. Using this model, we selected the top 131 designs and similarly folded these on the AlphaFold Server and computed pDockQ2 scores, yielding a median pDockQ2 of 0.13 and 33 with pDockQ2 > 0.2 over this second set.

To identify specific interactions that explain the pDockQ2 scores, for each complex, we computed PLiP (v3.0.0) interaction features. Since the Pup2 sequence is invariant, for each Pup2 residue and 8 interaction types [hydrophobic_packing, pi_stacking, and hbond_pdon, hbond_ldon, saltbridge_pneg, saltbridge_lneg, and pication_pdon, picaton_ldon] we counted the number of interactions with any residue of Poc4, where for the purposes of PLiP, Pup2 is the "protein" and Poc4 is the "ligand", yielding 77 non-zero features. Then we trained an H2O AutoML regression-ML model. The best performing model over 5-fold cross-validation by RMSD was a Gradient Boosted Model, with an RMSD of 0.041. To identify important features, using the fit model, we computed Shapley Additive exPlanation features using the TreeSHAP algorithm from H2O.

Growth assays

Spot assays were performed by growing fungal cells overnight in liquid YPD media at 30 °C with constant shaking. Cell culture density was normalized to an OD_{600} of 1 in sterile water and diluted 10-fold. 5 μL of each dilution was spotted onto YPD plates and incubated at 30 °C, 37 °C, or 42 °C for 2 days before imaging on a BioRad gel dock. Each plate image is representative of at least 3 biological replicates.

Kinetic growth assays were performed by growing fungal cells overnight in 200 μL of YPD in a 96-well plate at 30 °C, then subcultured into fresh media using a sterile pinner. Growth curves were performed on a BioTek LogPhase600 Plate Reader at the indicated temperatures for 24 h with shaking at 500 rpm and OD_{600} measurements taken every 10 min. A minimum of 4 biological replicates, each with 3 technical replicates, was performed per strain per growth condition. Growth statistics (rate, carrying capacity, and AUC) were calculated using the growthcurver R package. Growth curves were plotted in GraphPad Prism v11.

Minimum inhibitory concentration (MIC) assays

Standard compound sensitivity testing was evaluated by broth microdilution MIC testing in 96-well, flat-bottom microtiter plates. All compounds were dissolved in DMSO. Tested MG-132 (Cayman Chemical, 10012628) and bortezomib (Cayman Chemical, 10008822) concentrations ranged from 3.125 to 200 μM. Tested fluconazole (Cayman Chemical, 11594), caspofungin (GoldBio, C-520-10), and amphotericin B (GoldBio, A-560-250), concentrations ranged from 0.04–64 μg/mL, 0.08–80 ng/mL, and 0.04–64 μg/mL, respectively. Assays were set up in a total volume of 200 μL of YPD per well with 2-fold serial dilutions of the test compound. Following inoculation, plates were covered and incubated statically at the indicated temperature. Growth was quantified by measuring OD_{600} with a BioTek 800 TS Absorbance Reader at an endpoint of 24 h. Relative growth was calculated for each strain by dividing the optical density at a given concentration by the optical density of an untreated control on the same plate. All strains were assessed with 3 biological replicates in technical duplicates. Dose-response curves were plotted in GraphPad Prism v11.

Co-immunoprecipitation

Strains were inoculated in YPD and subcultured to an OD_{600} of 0.05 in 100 mL of fresh YPD. Cultures were grown for 6 h at 37 °C in 250 mL Erlenmeyer flasks with 150 rpm shaking until mid-log phase. Cells were collected by centrifugation at 1500 × *g*, washed in 1 mL sterile water, transferred to 1.5 mL screw-cap tubes, then flash frozen and stored at −80 °C.

Cell pellets were resuspended in 1 mL of lysis buffer (20 mM Tris, pH 7.5, 100 mM KCl, 5 mM MgCl, 20% glycerol, and 1× HALT protease inhibitor cocktail (Thermo Fisher Scientific, 87786) added fresh before use). The tube was filled with acid-washed glass beads until the beads were just below the meniscus at the top of the tube to limit foaming during bead beating. Cells were disrupted by bead beating three times for 4 min, with 7-min incubation on ice between cycles. Lysates were recovered by a stacked transfer and clarified by centrifugation at 18,000 × *g* for 10 min at 4 °C. Protein concentrations were determined by Bradford assay.

Anti-ALFA immunoprecipitations were performed with ALFA Selector PE magnetic agarose beads (NanoTag Biotechnologies, N1515). For each reaction, 20 μL of magnetic agarose bead slurry was washed 2 times in 1 mL ice-cold lysis buffer. Beads were collected by magnetic separation between each wash step. 1 mL of protein lysates, normalized to a total protein level of 2 mg and added to the washed beads. Reactions were incubated with end-over-end rotation for 2 h at 4 °C. The samples were then washed 3 times in 1 mL ice-cold lysis buffer, transferred to clean tubes, and washed again twice with 1 mL ice-cold lysis buffer, and one final time with 1 mL ice-cold Tris-buffered saline. The washed beads were then resuspended in 100 μL of Laemmli sample buffer, incubated at 95 °C for 5 min, shaking at 500 rpm. Supernatant was collected and subject to western immunoblotting on SDS-PAGE gels.

Protein extraction

Strains were inoculated overnight in YPD and subcultured to an OD_{600} of 0.1 in 10 mL of fresh media, incubated for 5 h with shaking at 30 °C. Following sub-culture, cells were harvested by centrifugation, then washed once with sterile water. Cells were resuspended in lysis buffer (50 mM HEPES, pH 7.4, 150 mM NaCl, 5 mM EDTA, 1% Triton X-100, 100 mM NaF, 1 mM PMSF, 1X HALT protease inhibitor cocktail (Thermo Scientific, 87786)) and lysed by bead beating. Following lysis, lysates were clarified by centrifugation, and protein concentration was determined by Bradford assay.

Western blotting

Protein levels were normalized to 25 μg and boiled in 1× Laemmli sample buffer for 5 min before loading on 4–20% SDS-PAGE gels (Bio-Rad, #4561094). Following separation, proteins were transferred to polyvinylidene difluoride (PVDF) membrane (Bio-Rad, #162-0177) for 15 min using a semi-dry transfer system. Blots were blocked with 5% bovine serum albumin (Sigma-Aldrich, A7030) in phosphate-buffered saline with 0.1% Tween-20 (PBS-T). The molecular weight was approximated by electrophoresis of marker proteins in either the PageRuler pre-stained protein ladder (Thermo Scientific, 26616) or PageRuler Plus pre-stained protein ladder (Thermo Scientific, 26619). ALFA epitopes were detected using camelid anti-ALFA Licor 800CW primary antibody (NanoTag Biotechnologies, N1502-Li800-L) at a 1:5000 dilution in blocking solution. FLAG epitopes were detected

using 1:1000 dilution rabbit anti-FLAG primary antibody (Sigma-Aldrich, F7425-.2MG). Blots were washed with PBS-T and incubated with 1:15000 goat anti-rabbit IgG IRDye-680RD (LI-COR Biosciences, 926-68071) or 1:15000 goat anti-rabbit IgG IRDye-800CW (LI-COR Biosciences, 926-32211). Ubiquitin epitopes were detected with 1:1000 dilution mouse anti-Ubiquitin primary antibody (Invitrogen, 13-1600). Blots were washed with PBS-T and incubated with 1:15000 dilution anti-mouse IgG IRDye-800CW secondary antibody (LI-COR Biosciences, 926-33210) before detecting signals using a LI-COR Odyssey scanner. Protein levels were normalized to total protein loading using Revert 700 total protein stain (LI-COR Biosciences, 926-11011). Proteasome α-subunits were detected using a primary antibody specific to a conserved peptide in the 20S alpha proteasome proteins Pre1, Pre2, Pre3, Pre5, Pre6, and Pre7 at a 1:3000 dilution in blocking solution (AbCam, antibody ID: AB_2171376). Following washing, blots were incubated with 1:15000 dilution goat anti-mouse IgG HRP secondary antibody (EDM Millipore, 12-348) before detecting signals using a ThermoFisher iBright 1500 imager. For quantification, band and total protein intensity were measured using ImageJ. Band intensities were normalized to the total protein signal in each lane. In each case, three biological replicates were quantified. All uncropped blots are provided in Source Data.

Statistics and reproducibility

Statistical analyses were carried out using GraphPad Prism v11. Data are presented as mean ± standard deviation from biological replicates. Unless otherwise noted, experiments were performed in at least three independent biological replicates with at least three technical replicates yielding similar results. All blots or images are representative of at least three independent experiments. Statistical significance among different groups was calculated using one-way ANOVA, with either Tukey or Dunnett's post hoc tests for multiple comparisons as appropriate. $^{*}p \le 0.05$; $^{**}p \le 0.01$; $^{***}p \le 0.001$; n.s., $p > 0.05$. Specific p-values are listed in Source Data.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data generated in this study are provided in this study are provided in the Supplementary Information and/or Source Data files. Strain used in this will be sent following standard material transfer agreements (MTAs) Source data are provided with this paper.

Code availability

All original code has been deposited at Zenodo at DOI: 10.5281/zenodo.19052665[77] and is publicly available as of the date of publication. The code is also developed openly at the GitHub repository (https://github.com/maomlab/Poc4).

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Acknowledgements

We thank the O'Meara Labs for helpful discussions. National Institutes of Health grant R35GM147894 (TRO), National Institutes of Health grant

U19AI181767 (TRO). National Institutes of Health grant T32GM149391 (JRR). National Institutes of Health grant R35GM151129 (MJO). National Institutes of Health grant 5F32CA261115 (MS). National Institutes of Health grant R35 GM118073 (PJW). Michigan Postdoctoral Pioneer Fellowship (MS)

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Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-71206-4.

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Peer review information *Nature Communications* thanks the anonymous reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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